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The Editors-in-Chief
Swarm and Evolutionary Computation

Dear Editors,

I submit the manuscript *"Does the learning matter? A pre-registered deconstruction of SARSA-guided information exchange in the MPHBS hybrid metaheuristic"* for consideration as a research article in *Swarm and Evolutionary Computation*.

Learned controllers are increasingly built into population-based metaheuristics, and their benefit is usually demonstrated by an ablation against a random or fixed control. The manuscript shows that this evidence can be misleading when the control is not matched to the learned variant in evaluation cost per iteration and in state updates. It does so for a recent, fully documented case: MPHBS (Flores and Olivares, *Knowledge-Based Systems* 352, 2026), a Honey Badger–Marine Predators hybrid whose information exchange is guided by a SARSA-style value table.

The main contributions are:

- an analytical proof, checked against the authors' public code and per-run results, that the published "matched" random mirror runs 21% (CEC2022) and 16% (FIR) fewer iterations at the same budget and, in one configuration, refreshes a population memory that MPHBS leaves unchanged;
- a pre-registered study with budget-identical controls (14,300 seed-paired runs on CEC2022 and eight FIR filter-design problems), analysed with a script frozen before the campaign: a random mediator performs at least as well as SARSA on both benchmarks, while the exchange step and the memory handling carry the performance;
- an exploratory extension to the 24 BBOB functions that replicates the absence of a learning effect;
- a definition of budget- and state-matched ablation controls and a protocol for constructing and verifying them.

I believe the work fits the journal's scope and its commitment to sound experimental methodology, to which the journal has contributed influential guidelines (Derrac et al., 2011; LaTorre et al., 2021). The study applies these principles to a learned control component and adds pre-registration and unit tests that prove the intended equivalences between variants. The findings are negative for the learned component but positive and practical for the field: they identify what actually drives the hybrid and give authors a checklist for ablating learned components.

All code, configurations, the frozen pre-registration, raw run data and analysis scripts are publicly available at `https://github.com/abinaya-g/la-kaie`. The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. The author declares no competing interests, and the research received no specific funding.

Thank you for considering this submission.

Yours sincerely,

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